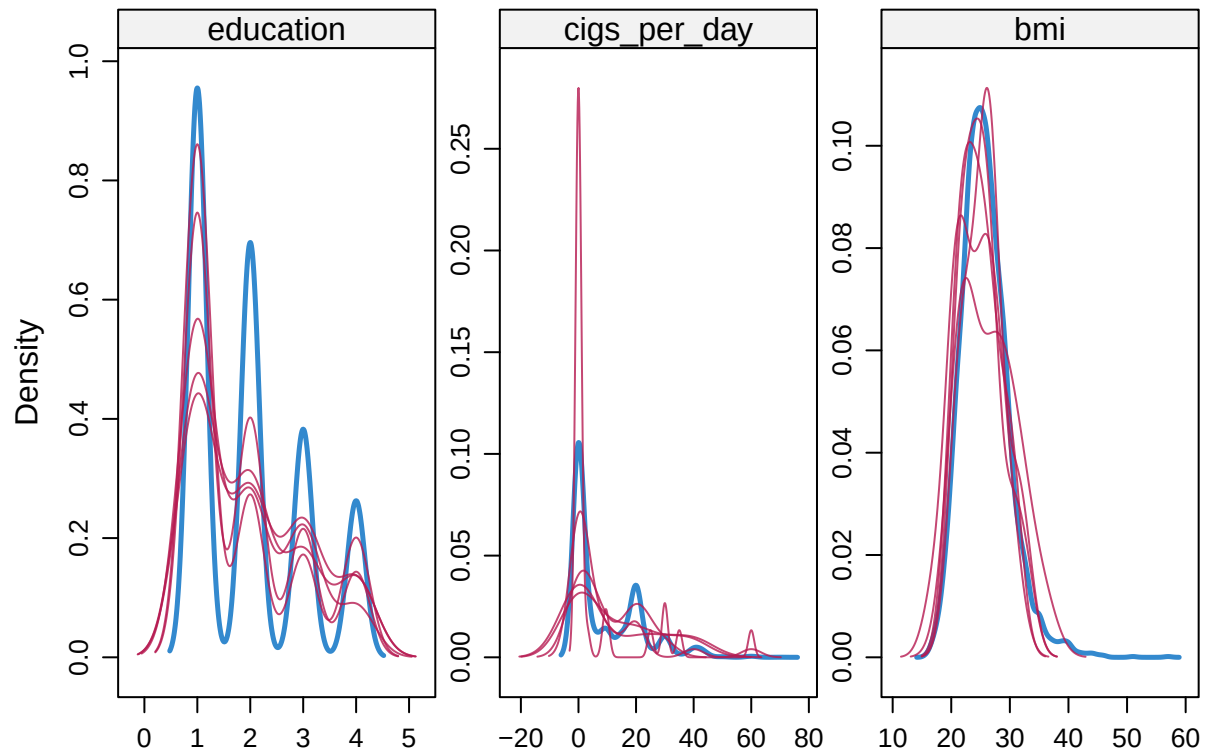


Assignment Two: Regression Analysis of Framingham Heart Study Data

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Handling Missing Data

In the Framingham Heart Study, there were significant missing data across a range of variables. Within the abbreviated dataset, variables with missing data were education, the number of cigarettes smoked per day, and respondents' BMI. To adequately handle these missing data, multivariate imputation by chained equations (MICE). Using the R `mice` package, the imputation model included all variables in the dataset (`age`, `sex`, `education`, `bmi`, and `cigarettes_per_day`) as predictors. Five imputed datasets were generated with 50 iterations each, using predictive mean matching for numerically stored variables. Convergence was then assessed by examining trace plots of imputed values. The first imputed dataset was used for all subsequent analyses. Density plots (see below) comparing observed and imputed values show good overlap, indicating that the imputations were plausible.



Distribution of Cigarettes per Day

As shown in the below histogram, the amount of cigarettes smoked per day has a stark non-normal distribution with a large right-skew and substantial zero-inflation. Therefore, using a Poisson regression is useful for understanding the impact of our independent variables. Poisson regressions do not have an r-squared value, and pseudo R-squared are controversial (Heinzel and Mittlböck 2003). While psudeo r-squared can have their place, this analysis opts for the use of Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) due to their prevention of overfitting (see Burnham and Anderson 2004). The distribution of cigarettes per day is displayed in the plot below.

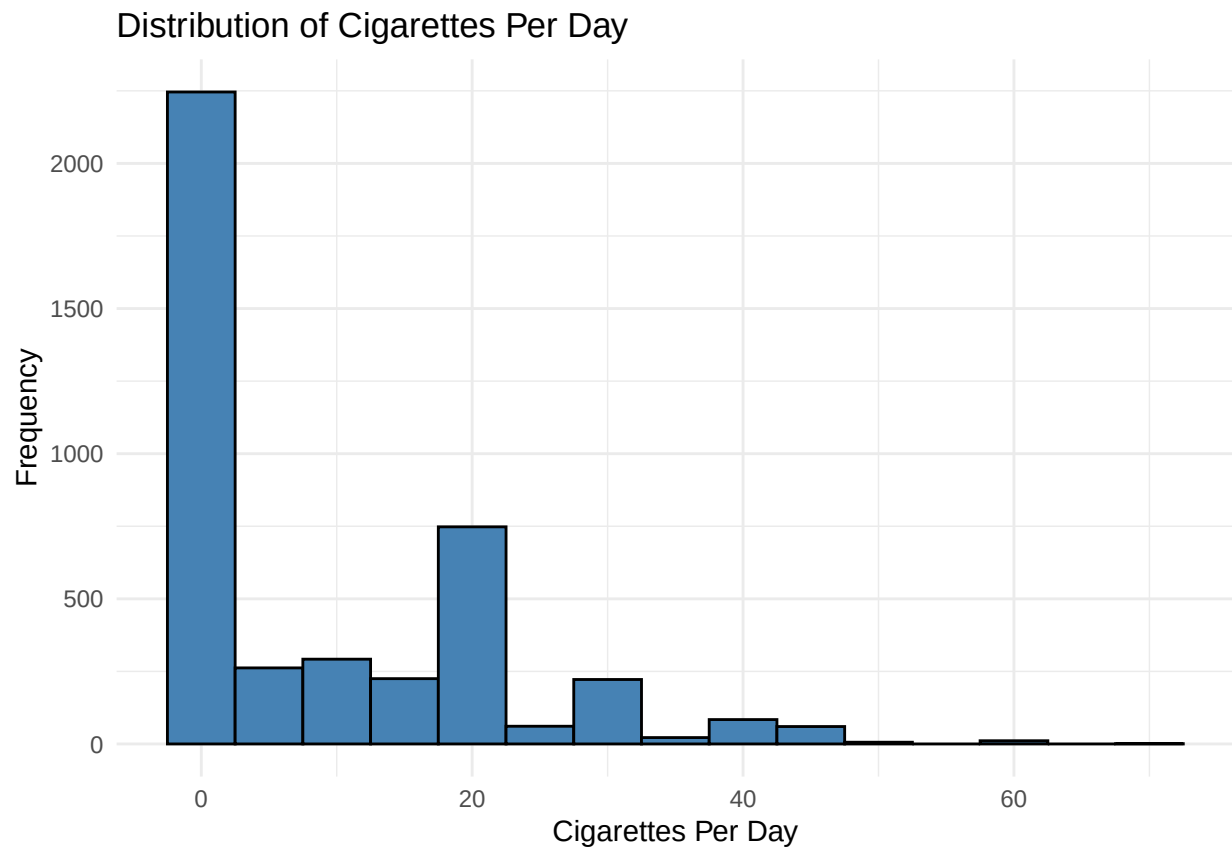


Table 1: Distribution Statistics of Cigarettes Per Day

Statistic	Value
Mean	9.02500
Median	0.00000
SD	11.95262
Min	0.00000
Max	70.00000
Zero Inflation (%)	50.94340

Regression Modelling of Cigarettes per Day

Model Fitting

Model Comparison

After running simple single predictor models on the number of cigarettes smoked per day, significant predictors across both linear and Poisson regression models are: **gender**, **age**, and **bmi**. Since these models are simple, and do not control for other variables, their explanatory power is relatively weak. However, we see that the variable **gender** has a positive association with males smoking more cigarettes than females. The variables **age** and **bmi** has a negative association, but have low explanatory value. Including these variables in a multivariate regression model would likely explain a greater share of variance. Detailed model results are presented in tables 5 and 6.

Table 2: Comparison of Linear and Poisson Regression Models

Model	Predictor	Linear Models					Poisson Models	
		Coefficient	Std. Error	P-value		R ² sq	AIC	BIC
Gender (LM)	Sex	7.6297	0.3519	0.0000	***	0.0998	32630.04	32649.10
Age (LM)	Age	-0.2682	0.0210	0.0000	***	0.0370	32916.14	32935.19
Education (LM)	Education	0.0646	0.1801	0.7199		0.0000	33075.93	33094.99
BMI (LM)	BMI	-0.2679	0.0447	0.0000	***	0.0084	33040.29	33059.34
Gender (Poisson)	Sex	2.3269	0.0106	0.0000	***	NA	71336.06	71348.77
Age (Poisson)	Age	0.9695	0.0006	0.0000	***	NA	75417.55	75430.26
Education (Poisson)	Education	1.0072	0.0050	0.1536		NA	77964.13	77976.83
BMI (Poisson)	BMI	0.9690	0.0013	0.0000	***	NA	77379.82	77392.52

Note:

Lower AIC/BIC indicates better fit. Signif. codes: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1

Table 3: Detailed Linear Model Results

	Gender	Age	Education	BMI
(Intercept)	5.750*** (0.231)	22.324*** (1.057)	8.897*** (0.400)	15.939*** (1.168)
sex	7.630*** (0.352)			
age		-0.268*** (0.021)		
education			0.065 (0.180)	
bmi				-0.268*** (0.045)
Num.Obs.	4240	4240	4240	4240
R2	0.100	0.037	0.000	0.008
R2 Adj.	0.100	0.037	0.000	0.008
AIC	32 630.0	32 916.1	33 075.9	33 040.3
BIC	32 649.1	32 935.2	33 095.0	33 059.3
Log.Lik.	-16 312.022	-16 455.068	-16 534.965	-16 517.143
F	470.098	162.901	0.129	35.907
RMSE	11.34	11.73	11.95	11.90

* p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors in parentheses. * p < 0.05, ** p < 0.01, *** p < 0.001

Table 4: Detailed Poisson Model Results (Incidence Rate Ratios)

	Gender	Age	Education	BMI
(Intercept)	5.750*** (0.049)	40.534*** (1.220)	8.898*** (0.099)	20.170*** (0.683)
sex	2.327*** (0.025)			
age		0.969*** (0.001)		
education			1.007 (0.005)	
bmi				0.969*** (0.001)
Num.Obs.	4240	4240	4240	4240
AIC	71 336.1	75 417.6	77 964.1	77 379.8
BIC	71 348.8	75 430.3	77 976.8	77 392.5
Log.Lik.	-35 666.031	-37 706.777	-38 980.063	-38 687.909
F	6315.746	2454.129	2.036	565.409
RMSE	11.34	11.75	11.95	11.90

* p < 0.05, ** p < 0.01, *** p < 0.001

Coefficients are incidence rate ratios. SE in parentheses. * p < 0.05, ** p < 0.01, *** p < 0.001

Gender-Specific Analysis

When subsetting the dataset by gender, we see that along with the previously significant variables, **education** becomes significant. This indicates that education as a variable while not significant alone, may be mediated by gender. This relationship can be untangled in further analyses using path models and factor analysis. As it stands, in simple regression models subsetted by gender, education becomes a predictor of decreased number of cigarettes smoked. Table 5 presents the results of a Poisson regression on these data subsets. All p-values are less than 0.001

Table 5: Gender-Specific Poisson Regression Results

Predictor	Gender	IRR	P-value	
Age	Male	0.9792	<0.0001	***
Education	Male	0.9688	<0.0001	***
BMI	Male	0.9702	<0.0001	***
Age	Female	0.9562	<0.0001	***
Education	Female	1.0779	<0.0001	***
BMI	Female	0.9355	<0.0001	***

Note:

IRR = Incidence Rate Ratio. Significance codes: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1